Name: Solutions (please print)

Signature: ______________________

ECE 2202 – Quiz 3

March 2, 2026

1. Show all work necessary to complete the problem. Use additional sheets of paper as needed.

2. Show all units in solutions and figures. Units in the quiz will be included between square brackets.

3. You will have 40 minutes to work on the quiz.

4. Write your ID number on every page that you show your work.

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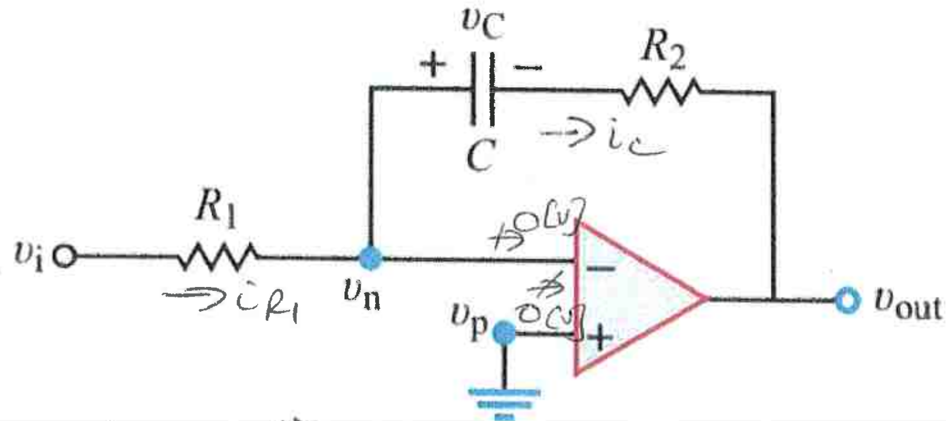
Problem 1/2 (8 pts)

Relate v_{out} to v_i in the circuit. Assume $v_c = 0$ at $t = 0$.



$i_{R_1} = i_c = \frac{v_i}{R_1}$

$i_c = C\frac{dv_c}{dt}$ or $v_c = \frac{1}{C}\int_0^t i_c\,dt$

KVL: $v_c + i_c R_2 + v_{out} = 0$

$\Rightarrow v_c + \left(\frac{v_i}{R_1}\right) R_2 + v_{out} = 0$

$= \frac{1}{C}\int_0^t \frac{v_i}{R_1}\,dt + v_i \frac{R_2}{R_1} + v_{out} = 0$

$$\boxed{v_{out} = -\frac{R_2}{R_1} v_i - \frac{1}{R_1 C}\int_0^t v_i\,dt}$$

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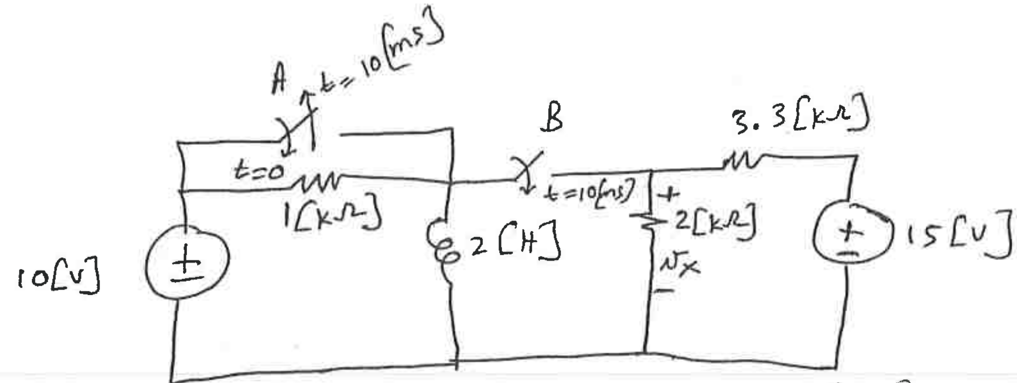
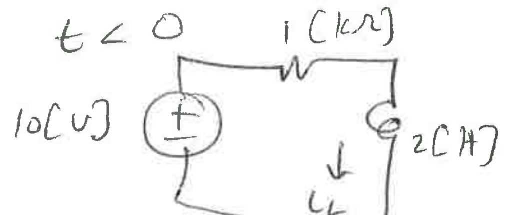
Problem 2 (12 pts)

In the circuit below, switch A and switch B are **open** for a long time. Then at t=0, switch A **closes**. Then at t=10 [ms], switch A **opens** and switch B **closes**. What is voltage $v_x(11\ [ms])$?

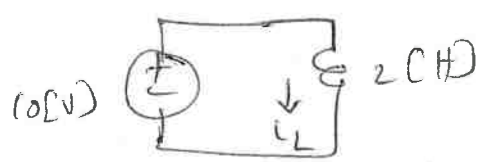


$t < 0$



$$i_L(0) = \frac{10[v]}{1(k\Omega)} = 0.01[A]$$

$0 \le t < 10[ms]$



$$10[v] = 2(H)\frac{di_L}{dt}$$

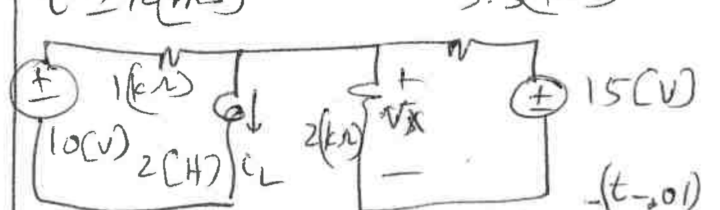
$$i_L = \frac{1}{2(H)}\int_0^t 10[v]dt + i_L(0)$$

$$= \frac{1}{2}10t + .01$$

$$i_L(10[ms]) = \frac{10}{2}(.01) + .01$$

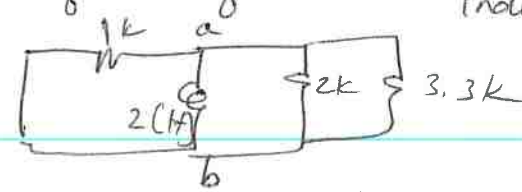
$$= 0.06[A]$$

$t \ge 10[ms]$



$$i_L(t) = i_{L\infty} + \left(i_L(10[ms]) - i_{L\infty}\right)e^{-\frac{(t-.01)}{\tau}}\ [A]$$

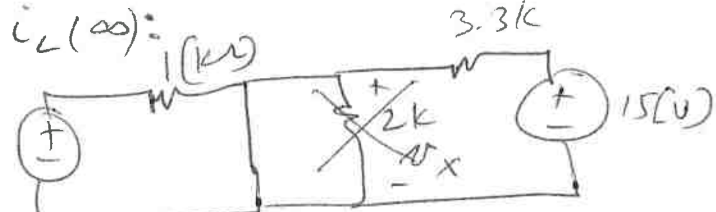
$\tau = \frac{L}{R_{eq}} = \frac{2(H)}{R_{eq}}$: R_{eq} is seen from a and b, by inductor:



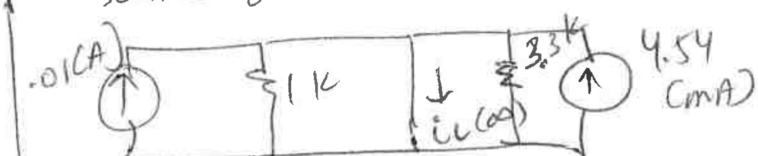
$$R_{eq} = 1k \| 2k \| 3.3k = 554.62\ (\Omega)$$

$$\tau = 0.0036\ [s]$$

$i_L(\infty)$:



Source transformation:



$$i_L(\infty) = 14.54\ (mA)$$

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Room for extra work

$$i_L(t) = 14.54 + \left(45.5 e^{-277.3 (t - .01[s])}\right) [mA]$$

$$v_x(t) = 2[H] \frac{d i_L(t)}{dt}$$

$$= 2[H] (.04545) (-277.3) e^{-277.3 (t - .01)} [V]$$

$$= -25.25 e^{-277.3 (t - .01)} [V]$$

$$v_x(11[ms]) = -19.105 [V]$$

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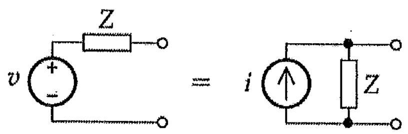
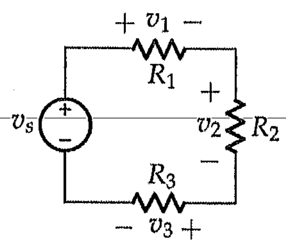
$$i = \frac{\mathrm{d}q}{\mathrm{d}t} \qquad v = \frac{\mathrm{d}w}{\mathrm{d}q} \qquad p = \frac{\mathrm{d}w}{\mathrm{d}t} = \frac{\mathrm{d}w}{\mathrm{d}q}\frac{\mathrm{d}q}{\mathrm{d}t} = iv \qquad R_1||R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 R_2}{R_1 + R_2}$$



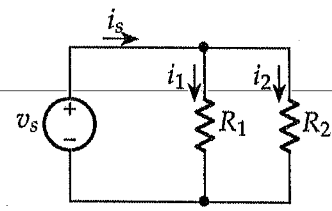
$$v = i \cdot Z$$
$$i = \frac{v}{Z}$$

passive sign convention:
$v = iR$
$p_{abs} = vi$
$p_{del} = -vi$
active sign convention:
$v = -iR$
$p_{del} = vi$
$p_{abs} = -vi$

$$p = i^2 R = \frac{v^2}{R}$$



$$v_1 = \frac{R_1}{R_1 + R_2 + R_3} v_s$$
$$v_2 = \frac{R_2}{R_1 + R_2 + R_3} v_s$$
$$v_3 = \frac{R_3}{R_1 + R_2 + R_3} v_s$$



$$i_1 = \frac{R_2}{R_1 + R_2} i_s$$
$$i_2 = \frac{R_1}{R_1 + R_2} i_s$$

Steps to find Thevenin/Norton Equivalent

- 1) Label terminals a and b
  - "Find the Thevenin or Norton equivalent circuit as seen by -----"
  - "Find the Thevenin or Norton equivalent circuit with respect to terminals a and b"
- 2) Find any components we can ignore (redraw if needed).
- 3) Decide which two of the three are we going to solve for: v_{oc}, i_{sc}, or R_{eq}
  - If there are dependent sources, solving for R_{eq} will require using a test source
- 4) v_{oc} circuit
  - Open across a and b, redraw, and solve for v_{oc}
- 5) i_{sc} circuit
  - Short across a and b, redraw, and solve for i_{sc}
- 6) R_{eq} circuit
  - Zero all your independent sources (short V-sources, open I-sources)
  - R_{eq} is the equivalent resistance as seen across a and b, or....
  - $R_{eq} = v_T/i_T$ where a test source is placed between a and b (active sign convention between v_T and i_T)
  - Redraw and solve for R_{eq}

$$i = C\frac{\mathrm{d}v}{\mathrm{d}t} \qquad v = L\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$p = Cv\frac{\mathrm{d}v}{\mathrm{d}t} \qquad p = Li\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$w = \frac{1}{2}Li^2 \qquad w = \frac{1}{2}Cv^2$$

$$v_C(t) = v_C(\infty) + [v_C(T_0) - v_C(\infty)]\, e^{-(t-T_0)/\tau}$$
$$i_L(t) = i_L(\infty) + [i_L(T_0) - i_L(\infty)]\, e^{-(t-T_0)/\tau}$$
(Switch action at $t = T_0$)

$$\omega = 2\pi f = \frac{2\pi}{T}$$
$$\frac{\phi}{360^\circ} = \frac{\Delta t}{T}$$
$$\sin(\theta) = \cos(\theta - 90^\circ)$$
$$\cos(\theta) = \sin(90^\circ - \theta)$$